

"How an LLM Works." $\Rightarrow$

Note: $\rightarrow$ Models cannot read full sentences like Human.

INPUT $\Rightarrow$ I am Human

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$\Rightarrow$ Not processed all at Once $\Leftarrow$

Instead it is processed Word by Word
Or
Token by token

Token

I = [1234]

am = [5968]

Human = [98745]

Tokenization

Breaking text into numbers.

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[1234, 5968, 9874]

Tokenization: It converts text into smaller chunks of tokens called Token.

Then, Each Token gets mapped to a number.

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Now, Numbers are still meaningless.

So, each token is converted into a Vector

List of Numbers

↓
which captures meaning.
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This uses → Word Embedding Idea.



I ["0.12, -0.44, ..."]

am = [0.87, 0.22, -0.11, ...]

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Now, the model understands the relationship

for eg. → cat is related to 'animal' or 'dog'

not 'car'.

→ photosynthesis relates to
plants, 'sunlight'

→ teacher is closer to 'School'
than to "engine"

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Now order matters in languages

(English,
Hindi etc)

So, model adds **positional Encoding**.

"dog bites man" $\neq$ "man bites dog"

For this, we use "positional Encoding."

Transformer.

Now, the data is ready to be processed.

the real brain starts working.

Transformer architecture.

This contains multiple layers
(Sometimes more than 100)

Each layer in transformer processes the input
again and again to refine understanding.

Self-Attention (Core Magic).



"Which words are important for understanding this word?"

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Eg → Explain photosynthesis simply.



When model processes photosynthesis.

Now it asks:

↳ How relevant is "Explain" to me?



↳ How relevant is "simply" to me?

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- * task
- * tone
- * grammar
- * meaning
- * relationship
- etc.

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